

Prit Mhala

pmhala@cs.stonybrook.edu | +1 (934) 223 0784 | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Stony Brook University, NY, USA Master of Science (MS) in Computer Science	Aug 2025 – May 2027 (GPA: 3.7/4)
Indian Institute of Information Technology Pune, India Bachelor of Technology (B.Tech.) in Computer Science and Engineering	Nov 2021 - Jul 2025 (GPA: 8.93/10)

EXPERIENCE

Deutsche Telekom Digital Labs, Gurugram, India	Jan 2025 – Jul 2025
- Developed and deployed a GDPR-compliant multilingual PII anonymization system as an AI Engineering Intern on the OneAI team, supporting enterprise chatbots <i>askT</i> and <i>Frag Magenta</i> across 7 European NATCOs, anonymizing 30+ PII entity types and serving over 2 million conversations across Europe using Git-based CI/CD workflows. - Designed a plug-and-play, real-time anonymization pipeline using Microsoft Presidio, spaCy, Flair, and regex-based rules, forming an ensemble NER architecture that achieved over 90% value-level accuracy. - Optimized production performance by scaling anonymization workflows on Spark clusters for chatbot log analytics, generating synthetic multilingual edge cases using Qwen 2.5 and Phi-4, validating models on AWS SageMaker with NVIDIA H100 GPUs, and enhancing chatbot safety and insights through NeMo Guardrails and NATCO-specific topic modeling.	

PROJECTS

MCP Multi-Agent Trading Simulation	[GitHub]
- Engineered an autonomous AI trading floor using the OpenAI Agents SDK with 4 GPT-4o-mini-powered agents, each embodying a distinct investment strategy (Value, Macro, Risk Parity, Disruptive Innovation), autonomously researching and executing real-time stock trades. - Architected a Model Context Protocol (MCP) infrastructure connecting each agent to 11 MCP servers for account management, real-time market data via Polygon.io, Brave Search-based financial research, LibSQL persistent memory, and Pushover push notifications with SQLite persistence. - Developed a real-time Gradio monitoring dashboard with Plotly charts displaying portfolio performance, transaction history with trade rationale, and live agent trace logs built on a custom asyncio tracing system for full decision-making transparency.	
AI Research Archaeologist: Genealogy Mapper	[GitHub]
- Built a full-stack research genealogy tool using a FastAPI backend and React frontend with Google Gemini 3's long-context reasoning and multimodal capabilities to perform deep content analysis of scientific papers, tracing idea evolution, identifying direct ancestors, and discovering missing links beyond standard citation tracking. - Designed an interactive genealogy graph visualization using React Flow and TailwindCSS, enabling researchers to explore paper lineage as a family tree with checkpoint-based session save/restore functionality. - Integrated LangChain-based agentic workflows with the Semantic Scholar API and Serper API for metadata retrieval and fallback web search, along with a conversational chatbot for querying the entire research genealogy.	
High-Performance 2PC-Sharded KV-Store	[GitHub]
- Developed a distributed sharded Key-Value store in Python integrating Multi-Paxos for intra-shard consensus replication and Two-Phase Commit (2PC) for atomic cross-shard transactions, ensuring strong linearizability and fault tolerance. - Implemented optimistic pipelining via asyncio to overcome traditional 2PC latency bottlenecks, achieving up to 50x throughput gains and handling 3,000+ concurrent transactions with fine-grained Two-Phase Locking (2PL). - Designed dynamic resharding for automatic load balancing across shards, with automated state transfer, log compaction, and periodic checkpointing for rapid recovery and scalable distributed operation.	

PUBLICATIONS

- **Published Research:** Proposed a deep learning algorithm for potato leaf disease classification achieving 81.31% accuracy in *Expert Systems with Applications (Impact Factor: 7.5)* . [\[Code\]](#) [\[Paper DOI\]](#)

TECHNICAL SKILLS AND RELEVANT COURSEWORK

Programming	Python, C, C++, Java, SQL, NoSQL (PostgreSQL, MongoDB, SQLite, LibSQL), JavaScript, TypeScript, HTML, CSS
Web & APIs	RESTful APIs, FastAPI, Node.js, Next.js, React.js, Angular, Vue.js, Flask, Gradio, Streamlit, React Flow, TailwindCSS, XML, JSON
AI/ML & Data	PyTorch, TensorFlow, CrewAI, AutoGen, LangGraph, LangChain, OpenAI Agents SDK, MCP, Hugging Face (Transformers, Diffusers), Google Gemini API, Amazon Bedrock, OpenCV, Plotly, Matplotlib, RAG, LlamaIndex, Prompt Engineering, Multi-Agent Systems, Agentic Workflows, Diffusion Models, Foundation Models, Gaussian Splatting, NeRF, pandas, NumPy
MLOps & Cloud	Docker, Kubernetes, Git, CI/CD, GitHub Actions, Terraform, AWS (SageMaker, Lambda, S3, CloudFront, SQS, App Runner, API Gateway, Route 53, Aurora Serverless, Bedrock AgentCore, Redshift), Azure, GCP (Vertex AI, BigQuery), IBM Cloud, Vercel, Weights & Biases, LangFuse, Databricks, Apache Spark, Apache Airflow, Apache Kafka, Looker, Grafana
Systems & Protocols	Distributed Systems, Multi-Paxos, Two-Phase Commit (2PC), Two-Phase Locking, asyncio, Concurrency, Fault Tolerance, Dynamic Resharding, ETL Pipelines, Data Warehousing